

Institute of Actuaries of India

Subject ST2 – Life Insurance

May 2014 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1 :

- Since the SCR is defined as a percentage of the sum at risk, the impact of change will depend on the sum at risk which is the difference between the sum assured and statutory reserves.
- The sum at risk of term assurance should be higher than traditional endowment products.
- Term assurance is 40% of the new business which would imply that as a percentage of sum assured, it would be much higher.
- Therefore on a portfolio basis the negative impact of term assurance will clearly outweigh any positive impact from the traditional endowment business but it also depends on the extent of the change.
- The impact will depend on the way the cost of capital is allowed to calculate profitability of the products. If the difference between the risk discount rate and the earned rate is not very high the impact on profitability will be lower.
- The impact on the existing traditional endowment business would also depend on how close the existing portfolio is to maturity. If the duration is close to maturity, the SCR requirement will be low and hence, the impact will be low and vice-versa.
- If there is an increase in SCR, it should not be an immediate concern since the company is a large insurance company and is expected to have sufficient free assets.
- The embedded value of the company will increase/decrease depending on the decrease/increase in the SCR.
- The company should check for the profit margins under term assurance for its new business, if still sustainable.
- If not, whether it requires re-pricing or new products.
- Under traditional endowment, the company may decide to reduce the premium rates to gain competitive edge.
- Though the impact on premium rates isn't likely to be much since the SAR under traditional endowment should be relatively small
- Most likely, the premium rates will remain unchanged.
- The SCR would increase exponentially going forward if the new business mix is expected to continue its current trend.
- Need to check if the company would be having sufficient free assets to fund this increase in capital requirement in the future.
- This would require model office projections to project the expected capital requirements and sensitivities using different product/business mix.
- The company may decide to limit the term assurance volumes if capital becomes a concern in future; not a popular choice.
- The company may take out reinsurance which would reduce the SAR and hence the SCR. This would require a cost-benefit analysis.
- The company may introduce other product types (eg: unit-linked, pensions, etc.) where the SCR requirement maybe less so that the overall SCR remains within acceptable limits.

[Total Marks -9]

Solution 2 : i)

- The expenses associated with the level of underwriting proposed. Underwriting involves a number of costs, such as salary of underwriter and medical reports, which will increase the overall costs to the company.
- The extent, and in particular the financial significance, of any potential anti-selection risk. If anti-selection is going to occur, it will present itself to the insurer with the least underwriting requirements within the peer group offering that product. Therefore, offering the same product with less underwriting may mean that an insurer attracts a disproportionate share of the anti-selection risks.
- The interaction between the level of underwriting and the potential level of sales – the greater the level of underwriting the lower the mortality costs but the expenses will rise. Also, people may be more inclined to take out contracts where there is a low level of underwriting.
- Claims underwriting may deter people from taking up a contract, due to the uncertainty as to whether a claim may be accepted.
- The effectiveness of the proposed underwriting – benefit exclusions may be difficult to police, or limiting medical evidence may make it difficult for staff to underwrite effectively. Non-disclosure also makes underwriting less effective.
- More detailed the underwriting, greater the homogenization of risk that may be achieved.
- The impact of regulation, which might constrain the level and/or type of underwriting that is permitted, eg the use of genetic testing.
- The interaction between underwriting level and terms offered by the company's reinsurers.
- How to vary underwriting criteria by age, sum assured, target market and other factors.
- Financial underwriting – depending upon the target market, the proof needed for financial insurability

[6]

ii)

- The NML of Rs.7.5 million may not be kept same across all ages, instead may be lowered for higher age groups, for example, 36-45 and 46+.
- Decrease the allowed ratio of the chosen SA to annual income to limit the risk.
- The level of disclosure required in the proposal form may be further increased.
- Only allow customers with a minimum annual income, say above Rs. 0.5 million
- Set min education qualifications as criteria for choosing, for example, graduate and above.
- Only proposals from top 50 urban cities be allowed.
- Only standard income and age proof be allowed.
- Few of the NML cases may be picked for medical tests randomly.
- Few of the deaths from the NML cases may be investigated and checked against the disclosure made by the customer during policy issuance.

[4]

(iii)

- The company may have been very conservative in setting its internet mortality assumption and hence may have used a heavier mortality assumption than has been experienced in practice.
- It may have done this due to the inexperience in the internet segment and hence wanted to build margins into its mortality assumption.
- The company may have wanted to control the amount of business it wrote through the internet due to uncertainty regarding the experience of the portfolio.
- It may have deliberately overpriced by assuming a heavier mortality assumption to ensure that it did not write more business than it wanted to.
- The actual mortality improvement experienced over the past years may have been greater than allowed in pricing.
- The company may have subsequently followed a stricter underwriting practice than assumed during the pricing stage leading to rejection of the most substandard lives.
- Or, the business mix is more favourable than expected during product pricing, for example, higher mix of medical lives, younger ages, non-smokers, etc.
- The company may have been more successful than anticipated at targeting the intended groups (income levels, etc.), with take up from these groups being higher than expected leading to overall lighter mortality amongst the internet policyholders than expected in the pricing basis.

[4]

[Total Marks-14]

Solution 3 : i)

- In order to minimize risk, a company should select investments that are appropriate to the nature, term and currency of the liabilities.
- The investments should also be selected so as to maximize the overall return on the assets, where overall return includes both investment income and capital gains.
- The extent to which the appropriate investments referred to above may be departed from in order to maximize the overall return will depend, inter alia, on the extent of the company's free assets and the company's appetite for risk.

[2]

ii)

- In formulating the investment strategy, the company needs to identify and allow for the risks with respect to asset liability mismatch.
- Since this is a traditional par endowment product, the assets backing it are likely to be invested in fixed-interest securities and equity.
- The main issue is that this is a regular premium contract, so the company doesn't know at what rates the future premiums can be invested.

- The risk is if interest rate reduces in early policy years, the value of liabilities may increase faster than the value of existing assets.
- The future investment income arising also needs to be reinvested on rates which are unknown at the product development stage.
- This risk gets reduced as the interest risk may be reflected in the future bonus rates subject to PRE and TCF considerations.
- The fixed interest securities are likely to back the guaranteed benefits and accrued bonus.
- The guaranteed surrender value is quite high in the initial years. So, if the interest rates are high, customer may surrender the policy and the company may incur a loss since the asset values will be depressed.
- Since the surrenders are uncertain, matching the surrender payouts is extremely difficult.
- The company may hence consider investing a proportion in short-term assets, though this will impact the overall return on the backing assets.
- If certain investments are to be made in equity and corporate bonds, then volatility and default risk will make the matching even more difficult.
- The company may hence restrict investments to government securities and high-rated corporate bonds , with equity investment.
- The company should also allow for the default risk if corporate bonds are required.
- The company should consider the use of derivatives and zero-coupon bonds, subject to availability and regulations.
- Going forward, the liabilities should allow for appropriate margins so that the assets held to cover liabilities are determined on prudent basis.
- Taking into account the above factors, a model office may be built. The liabilities representing the expected business mix and assets depicting the investment strategy should be projected forward on assumptions that represent expected future experience.
- The interest rate assumption should be set based on the current yields on the fixed-interest securities of appropriate duration and return on equity.
- In doing so, the company should also take a view of a future reduction in yields.
- For the assets, stochastic investment models can be incorporated in order to project future investment income and changes in capital values.
- The projected liabilities and assets can then be valued at the end of each year of the projection on the company's supervisory basis.
- Using a stochastic investment model and simulation techniques, the above can be extended to produce a statistical distribution of the amounts available each year to cover the level of solvency capital required and also to check that the bonus rates do not vary drastically from period to the next.
- Repeat the exercise for different investment strategies till it finds the most appropriate in terms of the company's objective.
- Invest the assets in the same currency as the liabilities.

[10]**[Total Marks-12]**

Solution 4 :

- What is the credibility of the analysts' conclusion? Does the company and the industry overall share the same view?
- The company might have already taken some steps recently to minimize mis-selling and the new UL business volumes hence might not get impacted much.
- If the shift to traditional products is being anticipated because of mis-selling of UL products, how can the industry be comfortable that the same will not get repeated for these set of products. The company will need to analyze whether a change in product offering will solve the problem.
- What are the potential sales volumes of the traditional endowment products? Can they replace the existing UL new business? Can the volumes cover the development and fixed costs?
- What are the likely profit margins under traditional endowment products? The company would like to at least retain the same level of profitability as under UL business.
- What are the capital requirements for writing traditional endowment products? The company needs to check if it has sufficient capital to fund the new business strain as well as for the development costs, etc.
- Will the target market remain the same as for the other products, specially the UL business? If not, whether the company needs to have a different channel and the associated costs involved. It may need to tie-up with some partners.
- The current sales force is selling UL and protection business. Hence, they may not be capable enough to sell traditional endowment products. The company therefore needs to consider the level of training and hence the investment in it.
- Will the new product suite impact the productivity of the sales force? If yes, then the company will need to re-look at the business model as the new business productivity will impact the expenses.
- Need to check the system compatibility. The system requirements, particularly for the with-profits business, will be different to the current products and would require system development. What is the cost of such development?
- Further, the company might not have the technical expertise to develop such products and other issues regarding management of the traditional endowment products. For example, pricing, declaration of bonuses, etc. Might need to seek consultant's advice and hence the cost.
- The risks involved with traditional endowment products are different to the UL and protection business. What are the risks and whether the company is comfortable in writing such risks? For example, interest rate risk is high under traditional without-profit products and the competition might require the interest rate guarantee to be high.
- ALM may not be a big concern for UL business as the company only has to bother about the non-unit liabilities. However in the traditional products, ALM becomes very significant. The company will have to invest in building this capability.
- What are the regulatory requirements for this product suite? For example, minimum benefits to be offered, requirement of a separate fund, establishing with-profits committee, etc. as well as the taxation rules.
- What are the competitors' products in the market and how are they performing? Are they successful? The products should be competitive enough to attract volumes.

- With-profits business is also driven by the past bonus rates and the investment record, which the company does not have. This will specially be the issue with brokers.
- The policyholder behavior of unit linked products is largely linked with the stock market and hence persistency is very unpredictable. This may not be the case for traditional products.

[Total Marks-10]

Solution 5 :

i) Risks associated with the product and mitigation

A. System risk

If the company has never sold with-profits business before, then the company needs to check whether appropriate systems are build-in to identify the various components of with-profits portfolio (such as premium income, investment income) and the various sources of surplus (investment, mortality expense etc.)

B. Competition/ Market risk

- a. If company has never sold or not actively sold with-profits business, risk that there is no benchmark available to policyholders how the bonus rates would be
- b. Policy though may look at investment performance of company unit-linked funds for comparison purposes
- c. Thus, risk arises that the bonuses declared are lower than offered by competitors
- d. Market risk also arises in terms of the distribution agents not able to understand the product properly and hence, a hindrance to sale
- e. Risk that annuities terms offered have higher guarantees
- f. Risk that surrender terms are competitive and/or guaranteed

C. Interest rate risk

- Moderate investment risk as this is being a with-profits product with guaranteed sum assured
- The investment risk is somewhat moderate and has leverage due to bonuses being non-guaranteed at outset and surrender terms being non-guaranteed.
- The risk is that the investment returns assumed in pricing/premium rate basis is not achieved
- If prudent investment return is assumed, there is a risk of product becoming uncompetitive

Possible mitigations would include

- offering low initial sum assured at the outset keeping in mind the competition
- making the contract unit-linked or accumulation type where investment return is passed at the end of the time period
- allowing deferred bonuses such as terminal bonus with lower regular bonuses

D. Mortality risk

- The risk is that there are higher deaths than expected leading to early payment of sum assured
- Depending upon the underwriting followed by the company, there is a risk of selection against the insurer in terms of attracting unhealthy lives

- Risk could also be due to the mix of business (in terms of age profile, lifestyle etc.) could be different than expected leading to different mortality experience
- If there is no reinsurance under the product, the risk is further elevated
- The risk is somewhat reduced since the mortality risk is shared with the policyholders
- Also, given this being savings contract, the risk is present in early years when sum at risk is high

Possible mitigations would include

- Offering low sum assured on deaths or in early years
- Making underwriting stringent or restriction entry age/maturity age

E. Longevity risk

- The risk is not significant as the annuity rates are determined at the point of maturity
- However, there is an implicit guarantee on annuity rates as annual annuity amount cannot be less than 3% of maturity value;
- However, the guarantee is not significant unless the vesting age is too low, (say starts as age 30 years) where the guarantee may bite depending on interest rates and average life expectancy

F. Lapse risk

- Risk of higher than expected withdrawals to not able to recover initial expenses
- Risk of selective withdrawal
- However, given with-profits plan, the risk is not significant

Possible mitigations would include

- Making the profits lapse neutral to the extent possible – such as no value in case early years lapse and paying higher bonuses in later years

G. New business risk (mix and volumes)

- Mix risk can lead to different mortality or lapse experience
- Mix risk can also lead to different average case size than expected
- Higher volumes could lead to capital strain
- Lower volumes could lead to not able to recover fixed expenses

Possible mitigations would include

- Introducing minimum premium size to ensure business mix risk is reduced

H. Legal, regulatory or tax risks

- There could be legal risk in terms of policyholder complaints about PRE about unfair contract terms
- Regulatory risk can arise in terms of not meeting PRE in declaring bonuses
- Risk that tax rate can change in future which is differ to what has been assumed in pricing
- Or risk of change in the manner or rate policyholders are taxed

I. New business strain risk

- a. Given this being a savings product, expected to have higher reserves from day 1 leading to capital strain

Possible mitigations would include

- Introducing minimum premium size to reduce new business strain related to fixed expenses
- Change the product design to actuarially funded unit linked product

[Total Marks-15]

(ii)

(a) Investment return

This will be set bearing in mind

- The significance of the assumption for the profitability of the contract, which will depend on the level of reserves built up and the investment guarantees given
- Given this being a savings contract, the level of reserves are high
- Also, the basic sum assured is guaranteed with non-guaranteed bonuses which implies moderate investment guarantee
- The extent of investment guarantee will determine the type of asset in which the premiums from the contract will be invested.
- The more onerous the investment guarantee, the more cautious approach the company will adopt in asset selection
- The extent of reinvestment risk
- This become a key parameter and it being a regular premium contract where future premium needs to be invested at unknown investment rates
- The intended asset mix for the contract.
- The intended investment mix will be affected by the factors as above (significance of investment guarantee, level of reserves, extent of reinvestment risk etc.)
- The intended investment mix will also be affected by the level of free assets available to the company.
 - This is because the extent of matching necessary to control the investment risk will be less when there are more free assets
- The current investment return of the asset mix and the likely future return
- If a market-consistent approach is used, either deterministically or stochastically, then the expected investment return can be set as the risk-free rate, irrespective of the actual underlying assets held (this is the “risk neutral” calibration approach).
- The impact of taxation, if any on the investment return needs to be allowed for

(6)

(b) Mortality rates and mortality improvement factors

- The assumption to be used should reflect the expected future experience of the lives who will take the plan
- This will depend on
 - The target market – typically dependent on the distribution channel involved
 - The underwriting controls applied (or not applied)

- The expected change in the experience since the time of the last historical investigation to the point in time at which the assumption will on average apply.
 - Mortality rates and mortality improvement factors will be based on an adjustment to standard mortality table and standard improvement factors table respectively
 - The adjustment could be based on own experience for type of contract concerned or similar class of business or industry data or reinsurer's data
 - Consideration should be paid to the expected changes in rates over time as this could be a particular issue for annuities where increased longevity is a risk
 - If there are significant death benefits, then any expectation of increasing mortality (say, due to AIDS) should be included in basis
 - The significance of setting accurate assumption given mortality and longevity risk is shared with the policyholders
- (5)

(iii)**a) Factors before considering reinsurance**

- cost of the reinsurance
 - counterparty risks – increase number of reinsurers and/or use deposit back
 - legal risks
 - type of reinsurance arrangement
 - amount of reinsurance (choosing retention limits).
 - Regulatory requirements
 - Appetite of the shareholders to retain mortality risk or willingness to pass mortality profit
 - Risk management policy
- (2)

(b) Reinsurance premium on quota share risk premium basis

- Determine the sum at risk for each policy.
 - In this case sum at risk will be the sum assured plus vested bonuses payable on death
 - Less the amount of reserves held in respect of the policy
 - The reinsured sum at risk will be determined by multiplying the sum at risk with (1- quota share retention proportion)
 - The risk premium will be applied to the above based on the age attained by the policyholder at that point.
 - If the reinsurance premium is determined annually, then the age attained by the policyholder and hence, the risk premium applied to it will change annually
 - Also, the sum at risk proportion will change annually due to change in reserves
- (3)

(c) Suitability of reinsurance arrangement including alternatives

- Company retains the same percentage of sum at risk for each policy
- The sum at risk can be potentially high depending upon the duration of contract
- Especially in the early years when reserves are not built
- Which could be a concern, especially for large case sizes
- Given the company is medium sized
- An alternative is to have a mix of quota share and individual surplus, retaining a percentage of each policy upto a specified limit
- Other alternative is to go individual surplus basis where the company knows its retention amount for each policy rather than retention proportion
- Original terms reinsurance may not be a good option here and there is a savings component involved

(4)**[Total Marks-35]****Solution 6 :****Principles****A. Surrender value to take into account policyholders' reasonable expectations**

- There is no generally accepted definition of PRE, but they will be influenced by, for example, the past practice of a company and any literature issued by it.

Whether it fits well on this principle depends on:

- What is being offered under its existing products of similar nature
- What is being offered by industry/ competitors
- Whether it is consistent with what is communicated to policyholders in illustration or policy literature at point of sale

B. surrender value should at early durations, not appear too low compared with premiums paid, taking into account any projections given at new business stage

- this seems to some extent consistent with the premiums paid by the policyholder
- in the early first or even second policy year, the asset share may well be negative due to high initial expenses
- paying out surrender value at least equal to 70% of premiums paid seems quite high

C. Surrender value should at later durations, be consistent with projected maturity values

- Using the pricing interest rate at the discounting rate, this principle fits this well as it will run well into the maturity value
- However, need to know what basis is used for other parameters, such as expense, mortality or inflation

- Though these parameters may not have a significant impact as that of interest rate as this being a saving policy and at later durations, outstanding term is small
- D. Surrender value should not exceed earned asset share, in aggregate over a reasonable time period
- Does not fit well in the early years as it is subject to a minimum surrender value
 - Surrender value currently has no reference to earned asset share
 - If the asset share has not earned the pricing interest rate till date, there is a risk that surrender value is higher than asset share
 - Or, If the asset share has earned the interest rate higher than pricing interest rate, the surrender value may appear low but this being non-participating plan, interest rate earned over and above pricing interest rate should not be passed to policyholder.
 - However, in early, it does not fit well due to paid-up sum assured moves linearly with number of premiums paid whereas each early premium has larger contribution to sum assured due to having longer outstanding period till maturity to earn investment income
- E. Surrender value should not lead to selection against the insurer
- Does not fit well as the interest rate is guaranteed
 - If the interest rates in market is lower than the pricing interest rate, policyholders may tend to surrender as there is a higher value received
 - If surrender value is higher than earned asset share, this may lead to lapse and re-entry
- F. Surrender value should take into account new business disclosure and what is offered by competitors
- This is similar to PRE requirement. Whether it fits well depends on whether the surrender value calculation formula is consistent to what is disclosed and how does this compare to what is offered by competitors
- G. Surrender value should avoid frequent changes, unless dictated by financial conditions
- Fits well as it is guaranteed formula with no changes
- H. Surrender value should avoid any significant discontinuities by duration
- Fits well as it is guaranteed formula with no changes and do not have any discontinuities and it is a smooth progression to maturity value assuming all other assumptions used to calculate surrender value is same as that of pricing assumptions
- I. Surrender value should not be excessively complicated to calculate
- Fits well as the formula is simple
- J. Surrender value should be capable of clear documentation
- Fits well as the formula is simple as it is guaranteed formula with no changes

K. Treating surrendering and continuing policyholders equitably

- Fits well at the later years as the formula is uses pricing interest rate but discontinuity due to minimum guaranteed surrender value.

[Total Marks-10]

Solution 7 :

- The main objective for such incentives are the following:
 - Enhance top-line to improve the market share
 - Improve productivity of the sales guys which helps in covering the fixed costs which should lead to higher margins,
 - Drive a certain behavior from the sales team.
- The company would like to analyze that the overall profitability doesn't get impacted much.
- The incentives will bring down the profit per policy. However, with the increase in new business volumes, the total profit might remain similar or even higher than before.
- The first incentive is a flat increase in company's cost by 10% of FYC in respect of the top quartile. The sales personnel will try to bring in as much new business as possible to qualify for this incentive. To this extent, this incentive meets the objective of increase in new business.
- If the company gets an additional 10% of new business or more from the top quartile, this incentive will be cost neutral or positive.
- The second incentive is w.r.t. persistency. Since the only charge is FMC, better the persistency, higher will be the profits.
- But the company would like to first compare the actual persistency with the proposed 70%. The proposition only makes sense if the actual persistency is much lower than 70%.
- The company will expect that the increase in profit from better persistency will outweigh the 5% FYC cost. This needs to be worked out by the company.
- First and second incentive might offset each other. For example, in order to qualify for the 10% FYC, there might be cases of mis-selling which would then ultimately lead to poor persistency. This is more likely since the first incentive is higher than the second incentive.
- To avoid this, the company may introduce a clawback clause on first year commission.
- Or the first and second incentive can be clubbed together instead of being independent. Though clubbing them would make it less likely to achieve increase in NB sales and improve persistency.
- Acquisition expenses comprise of both fixed costs and variable costs. Similarly for the renewal costs. Higher the average case size, lower will be the fixed costs as percentage of premium and hence higher will be the profit. The third incentive therefore intends to remunerate such who bring in business with better profitability.
- Additional sales of business with higher average case size will result in higher average profit per policy. Providing incentive will bring the profit per policy down, but as long as the total profit for the company increases, after allowing for the increased sales, the result is acceptable.

- As a minimum, the revised terms should ensure that the business covers its marginal costs, and makes a positive contribution to overheads and profit. However, all of the business cannot be priced in this way, or the company will fail to cover all expenses.
- The company also needs to consider whether it has sufficient assets to fund the additional new business sales, if the incentive structure becomes successful.
- Improvements in persistency might also be achieved by offering higher renewal commission, rather than via incentives.
- The suggestion seems to have merit as long as the objectives are being met.

[Total Marks-10]
